Shawn Xiong

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Education

University of Waterloo — Waterloo, ON
Bachelor of Applied Science in Computer Engineering, Co-op — *Sept. 2026 – Present*

Technical Skills

Languages: Python, TypeScript, C++, Java
Frameworks & Libraries: React, Next.js, ROS 2, librosa, NumPy, py_trees, pytest, Tailwind CSS
Tools & Hardware: Git, GitHub, Supabase, Arduino, STM32, MPU-6050, Soldering

Projects

ToneFFX | Live Demo | GitHub | *Python, TypeScript, Next.js, Supabase* — July 2026

- Built a guitar tone comparison app that turns differences between user recordings and reference audio into actionable EQ, presence, and tone adjustments.
- Handled timing differences between performances with chroma-based dynamic time warping in librosa; filtered low-energy and low-similarity matches before comparing 7 frequency bands and spectral brightness.
- Enabled users to record audio in-browser, save equipment settings through Supabase authentication and persistence, and refine recommendations using their previous adjustment attempts.

Giraffe – Wearable Posture Monitor | *C++, Arduino Nano, MPU-6050* — July 2025

- Won the Theme Prize at vsHacks 2025 in a 3-person team; developed firmware combining accelerometer and gyroscope readings to estimate posture and trigger a buzzer when the wearer slouched.
- Enabled sensor data acquisition through **register-level MPU-6050 configuration** using its datasheet and bit manipulation; helped assemble the wearable through soldering and 3D-printed enclosure integration.

Experience

Waterloo Aerial Robotics Group (WARG) — Sept. 2026 – Present
Autonomy Subteam Member — *Python, ROS 2, py_trees*

- Enabled on-demand image capture by developing a **Python ROS 2 node** that publishes a camera frame with GPS coordinates and IMU data when triggered.
- Combined inputs from **3 sensor topics** into a single capture message, giving downstream autonomy components image, location, and motion data through one interface.
- Built bootcamp behavior-tree logic to sequence flight and image capture at waypoints, check for fresh frames, and handle timeouts so capture tasks do not wait indefinitely.
- Created a simulated camera with cyclic image playback and monotonic timestamps to exercise perception logic without live camera hardware; implemented clockwise waypoint ordering.

Solar-Powered Boat Prototype | *UBC Physics Olympics* — March 2025

- **Led 4 teammates** in prototyping a solar-powered boat for UBC Physics Olympics, integrating energy storage and motor-control circuitry under heavily constrained power availability.
- Designed circuitry to store solar-panel energy in capacitors for motor propulsion; selected an STM32 microcontroller for low-power operation and designed a 60-second activation delay.
- Prototyped the circuit on a breadboard and assembled it on perfboard, moving the design from an adjustable test setup to a soldered prototype.

STEAM Summer Camp Assistant — July 7–18, 2025
Delta Trinity Lutheran Church

- Introduced **30 young children** to basic coding through guided Scratch activities, breaking tasks into simple steps and supporting introductory math lessons.
- Supported daily classroom activities by preparing equipment and materials, assisting instructors, and helping children follow activity instructions.